

VendorIQ AI

Intelligent Vendor Invoice Analytics Platform

Project Report

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Technology Stack

Python • Pandas • Scikit-learn • SQLite • Streamlit • Joblib

Project Summary

A machine learning application that predicts freight costs and identifies vendor invoices requiring manual approval using historical procurement data.

Version

1.0

Date

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1. Project Overview

VendorIQ AI is a machine learning-based decision support application designed to assist finance and procurement teams in vendor invoice analysis. The application consists of two predictive modules:

- Freight Cost Prediction
- Invoice Risk Analysis

The system was developed using Python, Scikit-learn, SQLite, Joblib, and Streamlit.

2. Problem Statement

Manual invoice verification is time-consuming and often results in delayed approvals and increased operational costs.

The objective of this project is to develop an intelligent application capable of:

- Predicting freight costs from invoice information.
- Identifying invoices that require manual approval.
- Supporting procurement and finance decision-making.

3. Dataset

The project uses historical purchasing and vendor invoice data stored in a SQLite database. The main tables include:

- Vendor Invoice
- Purchases

Important features include:

- Invoice Amount
- Freight Cost
- Invoice Quantity
- Total Item Quantity
- Total Item Value

All monetary values are recorded in US Dollars (USD).

4. Methodology

The project follows a standard machine learning workflow.

1. Load data from SQLite
2. Data preprocessing
3. Feature engineering
4. Train-test split
5. Feature scaling
6. Model training
7. Model evaluation
8. Integration of trained machine learning models into a Streamlit application

5. Machine Learning Models

5.1. Freight Cost Prediction

The freight cost prediction task was formulated as a supervised regression problem, where the objective was to estimate the freight cost associated with a vendor invoice using the invoice amount as the input feature.

Three regression algorithms were implemented and evaluated:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Model	MAE	RMSE	R^2 (%)
Linear Regression	24.11	124.72	96.99
Decision Tree Regressor	32.97	150.31	95.63
Random Forest Regressor	26.13	134.79	96.48

Table 1: Performance comparison of regression models.

Among the evaluated models, Linear Regression achieved the lowest prediction error (MAE = 24.11, RMSE = 124.72) and the highest coefficient of determination ($R^2 = 96.99\%$). Consequently, the Linear Regression model was selected for integration into the VendorIQ AI application. The trained model was serialized using Joblib and loaded by the Streamlit application to generate freight cost predictions.

5.2. Invoice Risk Analysis

Invoice risk analysis was formulated as a binary classification problem to determine whether an invoice should proceed through the standard approval workflow or be flagged for manual review.

Three classification algorithms were evaluated:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Model performance was evaluated using Accuracy, Precision, Recall, and F1-score.

Model	Accuracy
Logistic Regression	83%
Decision Tree Classifier	86%
Random Forest Classifier	91%

Table 2: Accuracy comparison of classification models.

The Random Forest Classifier achieved the highest classification accuracy of approximately 91%, outperforming both Logistic Regression (83%) and the Decision Tree Classifier (86%). Based on these results, the Random Forest Classifier was selected for invoice risk prediction.

The trained classifier was serialized using Joblib and integrated into the Streamlit application, where it predicts whether an invoice should be approved automatically or routed for manual review.

6. Application Interface

6.1. Dashboard

 **VendorIQ AI**
Intelligent Vendor Invoice Analytics Platform
Forecast freight costs and identify high-risk vendor invoices using machine learning to support smarter finance and procurement decisions.

 **Freight Cost Predictor**
Estimate expected freight charges before invoice processing.

Invoice Amount (\$)
0.00 - +

Predict Freight Cost

 **Invoice Risk Analyzer**
Determine whether an invoice should be routed for manual approval.

Invoice Quantity
0 - +

Total Item Quantity
0 - +

Invoice Amount (\$)
0.00 - +

Total Item Value (\$)
0.00 - +

Freight Cost
0.00 - +

Analyze Invoice

Figure 1: VendorIQ AI Dashboard

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6.2. Freight Cost Prediction

**VendorIQ AI**
Intelligent Vendor Invoice Analytics Platform
Forecast freight costs and identify high-risk vendor invoices using machine learning to support smarter finance and procurement decisions.

**Freight Cost Predictor**
Estimate expected freight charges before invoice processing.

Invoice Amount (\$)
900.00

Predict Freight Cost

Prediction completed successfully.

Estimated Freight Cost
\$10.00

**Invoice Risk Analyzer**
Determine whether an invoice should be routed for manual approval.

Invoice Quantity
0

Total Item Quantity
0

Invoice Amount (\$)
0.00

Total Item Value (\$)
0.00

Freight Cost
0.00

Analyze Invoice

Figure 2: Freight Cost Prediction

6.3. Invoice Approved

**VendorIQ AI**
Intelligent Vendor Invoice Analytics Platform
Forecast freight costs and identify high-risk vendor invoices using machine learning to support smarter finance and procurement decisions.

**Freight Cost Predictor**
Estimate expected freight charges before invoice processing.

Invoice Amount (\$)
900.00

Predict Freight Cost

**Invoice Risk Analyzer**
Determine whether an invoice should be routed for manual approval.

Invoice Quantity
8

Total Item Quantity
8

Invoice Amount (\$)
543.00

Total Item Value (\$)
543.00

Freight Cost
7.00

Analyze Invoice

 Suitable for Standard Approval

Recommended Action
Proceed with the standard approval workflow.

Figure 3: Invoice Suitable for Standard Approval

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6.4. Invoice Flagged

The screenshot displays a web application interface with two main panels. The left panel, titled 'Freight Cost Predictor', contains a text input for 'Invoice Amount (\$)' with the value '900000.00' and a 'Predict Freight Cost' button. The right panel, titled 'Invoice Risk Analyzer', contains several input fields: 'Invoice Quantity' (8), 'Total Item Quantity' (7), 'Invoice Amount (\$)' (90000.00), 'Total Item Value (\$)' (900.00), and 'Freight Cost' (90.00). Below these fields is an 'Analyze Invoice' button. A red banner below the inputs states 'Manual Approval Recommended'. Underneath, a section titled 'Recommended Actions' lists four bullet points: 'Verify invoice values.', 'Review supporting documentation.', 'Validate vendor transaction history.', and 'Complete manual approval before payment.'

Figure 4: Invoice Requiring Manual Approval

7. Technologies Used

Technology	Purpose
Python	Programming Language
SQLite	Database
Pandas	Data Processing
Scikit-learn	Machine Learning
Joblib	Model Serialization
Streamlit	Web Application
Git	Version Control

Table 3: Technologies Used

8. Conclusion

VendorIQ AI demonstrates the practical application of supervised machine learning to support vendor invoice analysis through an interactive decision support application.

The application provides the following capabilities:

- Freight cost prediction using historical invoice data.
- Automated identification of invoices requiring manual review.

- An interactive web-based interface for model inference.
- Decision support for finance and procurement processes.

Although the application demonstrates the complete machine learning workflow from data preprocessing to model integration, several enhancements could be incorporated in future work. These include expanding the feature set using additional vendor and purchasing information, evaluating alternative machine learning algorithms, integrating the application with enterprise database systems, and deploying the application to a cloud platform for broader accessibility.